

Damage to the Left Precentral Gyrus Is Associated With Apraxia of Speech in Acute Stroke

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Background and Purpose—Apraxia of speech (AOS) is a motor speech disorder, which is clinically characterized by the combination of phonemic segmental changes and articulatory distortions. AOS has been believed to arise from impairment in motor speech planning/programming and differentiated from both aphasia and dysarthria. The brain regions associated with AOS are still a matter of debate. The aim of this study was to address this issue in a large number of consecutive acute ischemic stroke patients.

Methods—We retrospectively studied 136 patients with isolated nonlacunar infarcts in the left middle cerebral artery territory (70.5 ± 12.9 years old, 79 males). In accordance with speech and language assessments, the patients were classified into the following groups: pure form of AOS (pure AOS), AOS with aphasia (AOS-aphasia), and without AOS (non-AOS). Voxel-based lesion-symptom mapping analysis was performed on T2-weighted images or fluid-attenuated inversion recovery images. Using the Liebermeister method, group-wise comparisons were made between the all AOS (pure AOS plus AOS-aphasia) and non-AOS, pure AOS and non-AOS, AOS-aphasia and non-AOS, and pure AOS and AOS-aphasia groups.

Results—Of the 136 patients, 22 patients were diagnosed with AOS (7 patients with pure AOS and 15 patients with AOS-aphasia). The voxel-based lesion-symptom mapping analysis demonstrated that the brain regions associated with AOS were centered on the left precentral gyrus.

Conclusions—Damage to the left precentral gyrus is associated with AOS in acute to subacute stroke patients, suggesting a role of this brain region in motor speech production. (*Stroke*. 2016;47:31-36. DOI: 10.1161/STROKEAHA.115.010402.)

Key Words: aphasia ■ apraxia, articulatory ■ speech ■ stroke

After Paul Broca seminal work published in 1861, the center for speech production have been thought to be located in the anterior half of the language-dominant hemisphere.¹ Specifically, Broca considered that the left posterior inferior frontal gyrus plays a pivotal role in speech production based on the clinicopathological observations of patients who exhibited problems with articulation, word finding, repetition, production, and comprehension of complex grammatical structures, which is the syndrome now recognized as Broca's aphasia. However, the functional role of the left inferior frontal gyrus has been controversial because Broca patients had language deficits other than speech production impairment and their lesions extended beyond the inferior frontal gyrus.^{2,3}

Today, speech production is not considered as a single function but to comprise multiple distinct processes, including retrieval of verbal symbols, motor speech programming or planning, and end-stage execution of articulation.^{4,5} Apraxia of speech (AOS) is the term denoting impairment in motor speech programming/planning, and it is differentiated from aphasia and dysarthria. The clinical characteristics of AOS

include slow rate speech, articulatory distortions, articulatory groping, lengthened intersegment duration, and segmentation of syllables.⁶ As each of these speech features is observed not only in AOS but also in phonemic paraphasia or dysarthria, the diagnosis of AOS is made based on specific combinations of them.^{7,8} Although a considerable number of clinicoanatomic studies of AOS have been published,⁹⁻²³ few studies have attempted to clarify brain regions associated with AOS using neuroimaging analysis of relatively large samples. Consistent with Broca initial postulation, some of these studies concluded that AOS is attributable to lesions in the left posterior inferior frontal gyrus.^{2,18,19} In contrast, other studies suggested that AOS is associated with lesions located in areas other than the inferior frontal gyrus, such as the left anterior insula^{20,21} and the left precentral gyrus.^{22,23} The discrepancy observed in these studies may arise from the following factors: (1) functional recovery because of plasticity or compensation in chronic aphasia; (2) lesion overlapping procedures performed without a statistical comparison between patients with AOS and those without; and (3) employment of imaging modalities

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with low spatial resolution, such as computed tomography or perfusion-weighted images for lesion analysis. To eliminate these confounding factors as much as possible, we investigated the brain regions associated with AOS in a large number of patients with acute ischemic stroke using statistical analysis of voxel-based lesion mapping on T2-weighted image (T2WI) or fluid-attenuated inversion recovery (FLAIR) image.

Methods

Study Population

The subjects of this study were selected from 2146 consecutive patients with acute ischemic stroke who were admitted to Kohnan Hospital (Sendai, Miyagi, Japan) between April 2007 and March 2012. The patients were admitted to the hospital within 7 days after stroke onset. The clinical and investigative data prospectively collected in a standardized fashion were entered into the Kohnan Hospital Stroke Registry. A neurologist, neurosurgeon, or both examined all patients. Patients were also subjected to routine laboratory tests and computed tomography or magnetic resonance imaging (MRI). On the basis of the clinical and brain imaging findings, board-certified neurologists, who specialized in the care of patients with stroke, made a diagnosis of ischemic stroke. The severity of neurological deficits was evaluated using the National Institutes of Health Stroke Scale (NIHSS) score on admission.²⁴ The inclusion criteria for this study were as follows: (1) first-ever stroke onset, (2) isolated nonlacunar infarcts in the left middle cerebral artery territory verified on MRI, (3) right handed, (4) no previous history of dementia, and (5) neuropsychological evaluation by speech-language pathologists during his/her hospital stay. Patients with severely reduced spontaneous speech production were excluded because of the difficulty of evaluating motor speech abilities. Three cases were also excluded because of the inadequacy of brain imaging. In total, 136 patients (70.5±12.9 years old, 79 males) were included. The median NIHSS score of the patients on admission was 5 (2–10, interquartile range). This study was approved by the Kohnan Hospital Institutional Review Board.

Speech and Language Assessments

The presence or absence and the classification of motor speech impairment and aphasia were determined by speech-language pathologists in the following manner. First, patients' speech and language abilities were screened in 10-minute free conversation, repetition (10 words and 3 short sentences), and 20-item picture naming. Buccofacial praxis was assessed on volitional oral and facial movements, including coughing, clicking the tongue, licking the lips, and whistling. Patients who exhibited any speech abnormalities, word finding difficulty, or ≥1 errors on the repetition, naming, or buccofacial praxis underwent further assessments with the standard language test of aphasia to determine the diagnosis of speech and language disorders.²⁵ The standard language test of aphasia is a comprehensive test of language functions for adult Japanese speakers and comprises subtests for auditory word and sentence comprehension, object naming, word and sentence repetition, cartoon description, verbal fluency, reading aloud and comprehension, writing, and calculation.

Motor speech abilities were assessed by perceptual observations of free conversation (several tens of minutes) and the cartoon descriptions, repetition, and reading aloud subtests of the standard language test of aphasia. These assessments were qualitative and no quantitative assessment tools for motor speech were incorporated. The minimum diagnostic criteria for AOS were slow speech rate and distorted sound substitutions or additions.^{7,8} In addition, we took into account the presence of articulatory groping, lengthened intersegment duration, and segmentation of syllables. Patients were diagnosed as having aphasia when word finding difficulty, anomia, paraphasia, or comprehension impairment were observed. The diagnosis of aphasia did not rely on the presence of agrammatism because agrammatic speech is not salient features of aphasia in agglutinative languages, including Japanese.^{26,27}

On the basis of the presence or absence of AOS and aphasia, patients were classified into the following 3 groups: (1) pure AOS, in which patients had AOS but not aphasia; (2) AOS with aphasia, in which patients had both AOS and aphasia; and (3) non-AOS, in which patients did not have AOS regardless of the presence or absence of aphasia. The median interval between stroke onset and neuropsychological evaluation was 7 days (5–10).

Imaging Procedures

Lesion locations were determined using T2WI or FLAIR images, the acquisition of which occurred close in time to the neuropsychological evaluation. The intervals between neuropsychological evaluation and MRI acquisition and between onset and MRI acquisition were 3 days (1–6.75) and 9 days (7–12.75), respectively. All scans were performed on a 1.5-tesla unit (Signa Excite, GE Medical Systems, Milwaukee, WI). The following parameters were used for T2WI acquisition: repetition time, 3000 ms; echo time, 80 ms; matrix, 320×256; field of view, 22×22 mm; section thickness, 6 mm; and intersection gap, 2.0 mm. For FLAIR acquisition, the following parameters were used: repetition time, 8002 ms; echo time, 126 ms; inversion time, 2000 ms; matrix, 256×224; field of view, 22×22 mm; section thickness, 6 mm; and intersection gap, 2.0 mm.

A single investigator (R.I.), who was blind to the results of speech and language assessments, manually demarcated each patient's lesions on the original T2WI (118 patients) or FLAIR (18 patients) image using MRICron (<http://www.mccauslandcenter.sc.edu/mricron/mricron/>) with reference to the initial diffusion-weighted image. Original images and lesion masks were normalized to a standard T2WI or FLAIR template (<http://glahnrgroup.org/Members/anderson/flair-templates>) using a nonlinear transformation algorithm with lesion cost-function masking,²⁸ which is implemented in Statistical Parametric Mapping 8 (SPM8) software (The Wellcome Trust Center for Neuroimaging, The Institute of Neurology at University College London, London, United Kingdom). The resultant images were then resampled into 2-mm isotropic voxels. The precision of spatial normalization was ensured by visual inspection with reference to the standard templates.

Voxel-Based Lesion–Symptom Mapping

Voxel-based lesion–symptom mapping (VLSM) analyses were performed using nonparametric mapping (<http://www.mccauslandcenter.sc.edu/mricron/npm/>) and MRICron software.²⁹ Intergroup comparisons were made between the following groups: all AOS (pure AOS plus AOS-aphasia) and non-AOS, pure AOS and non-AOS, AOS-aphasia and non-AOS, and pure AOS and AOS-aphasia. The presence or absence of a lesion in a given voxel was compared between the 2 groups using the Lieberman test.²⁹ The voxels in which ≥3% of the subjects had lesions were included in the analysis. A 5% family-wise thresholding with 3000 permutations was used to correct for multiple comparisons. The automated anatomic labeling³⁰ and the Anatomy toolbox³¹ were used for anatomic localization. In addition to VLSM maps, we generated the lesion overlap maps to present the overall distribution of lesions for each group.

Statistical Analysis

The intergroup comparisons of age, initial NIHSS, time interval between onset and speech/language evaluation were made using the Kruskal–Wallis test. Chi-squared test was used to test intergroup differences in sex proportion, stroke subtypes and the presence or absence of dysarthria, word finding difficulty, and buccofacial apraxia. These analyses were performed using the JMP (SAS Institute Inc, Cary, NC) statistical software package. A value of $P < 0.05$ (2-sided) was considered to indicate a statistically significant difference.

Results

Of the 136 patients, 7 patients were diagnosed as pure AOS, 15 patients as AOS-aphasia, and 114 patients as non-AOS.

Table. Clinical Characteristics of Patients

Feature	Pure AOS (n=7)	AOS-aphasia (n=15)	Non-AOS (n=114)	P Value
Age, y, median (IQR)	64 (50–72)	69 (63–80)	73 (63.75–80)	0.2329
Male, No. (%)	5 (71)	7 (47)	67 (59)	0.5124
Initial NIHSS, median (IQR)	1 (1–10)	10 (2–21)	4.5 (2–9)	0.0504
Onset to language evaluation time, d, median (IQR)	6 (4–10)	6 (6–11)	7 (5–10)	0.9231
Stroke subtypes, No. (%)				
Cardioembolic stroke	1 (14)	6 (40)	46 (40)	0.2415
Large artery disease	2 (29)	7 (47)	40 (35)	
Others	4 (57)	2 (13)	28 (25)	
Dysarthria, No. (%)	0 (0)	0 (0)	17 (15)	0.1534
Word finding difficulty, No. (%)	0 (0)	14 (93)	51 (45)	<0.0001
Buccofacial apraxia, No. (%)	1 (14)	8 (53)	11 (10)	<0.0001
Aphasia subtypes, No. (%)				
Broca's aphasia	...	15 (100)	0 (0)	...
Wernicke's aphasia	...	0 (0)	32 (28)	...
Anomic aphasia	...	0 (0)	19 (17)	...
Transcortical sensory aphasia	...	0 (0)	16 (14)	...
Others	...	0 (0)	7 (6)	...
No aphasia	...	0 (0)	40 (35)	...

AOS indicates apraxia of speech; IQR, interquartile range; and NIHSS, National Institutes of Health Stroke Scale.

Demographic and clinical profiles about stroke and the results of speech and language assessments of each group are summarized in Table. The lesion overlapping maps for individual groups are shown in Figure 1.

The results of VLSM analyses are shown in Figures 2 and 3. The regions associated with all AOS (pure AOS plus AOS-aphasia) were centered on the posterior wall of the left precentral gyrus in the central sulcus ($Z \geq 2.838$, Figure 2A).

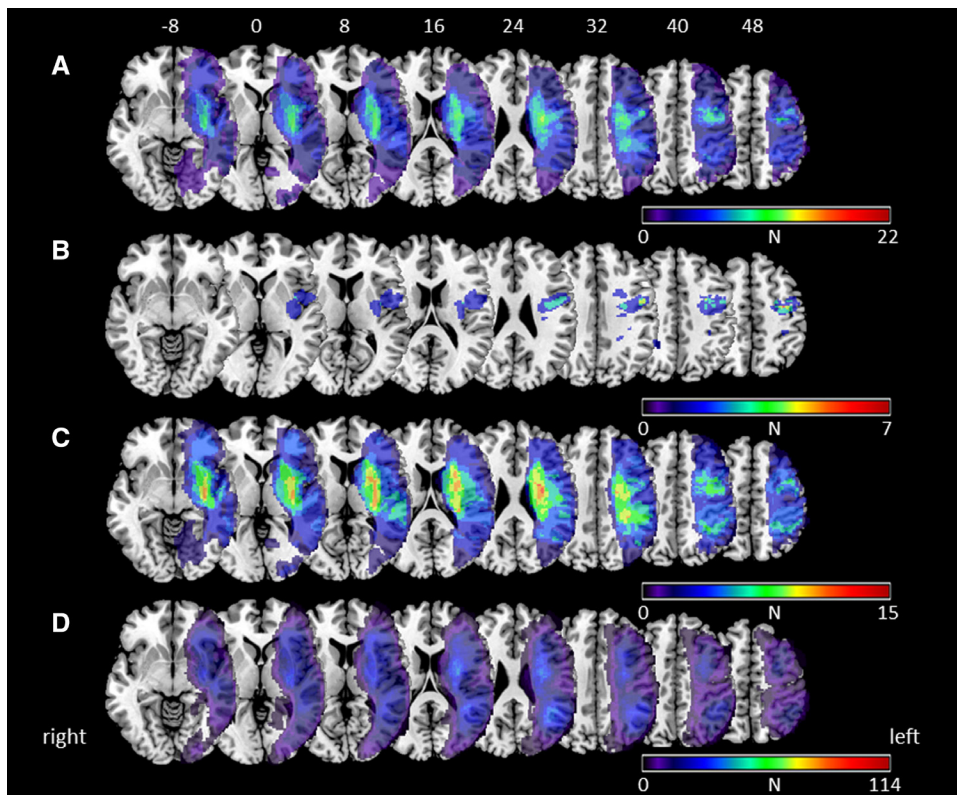


Figure 1. Lesion overlapping maps for individual patient groups. **A**, All apraxia of speech (AOS) patients (n=22). **B**, Patients with pure AOS (n=7). **C**, Patients with AOS-aphasia (n=15). **D**, Patients without AOS (n=114). Blue and red indicate the least and most overlaps, respectively.

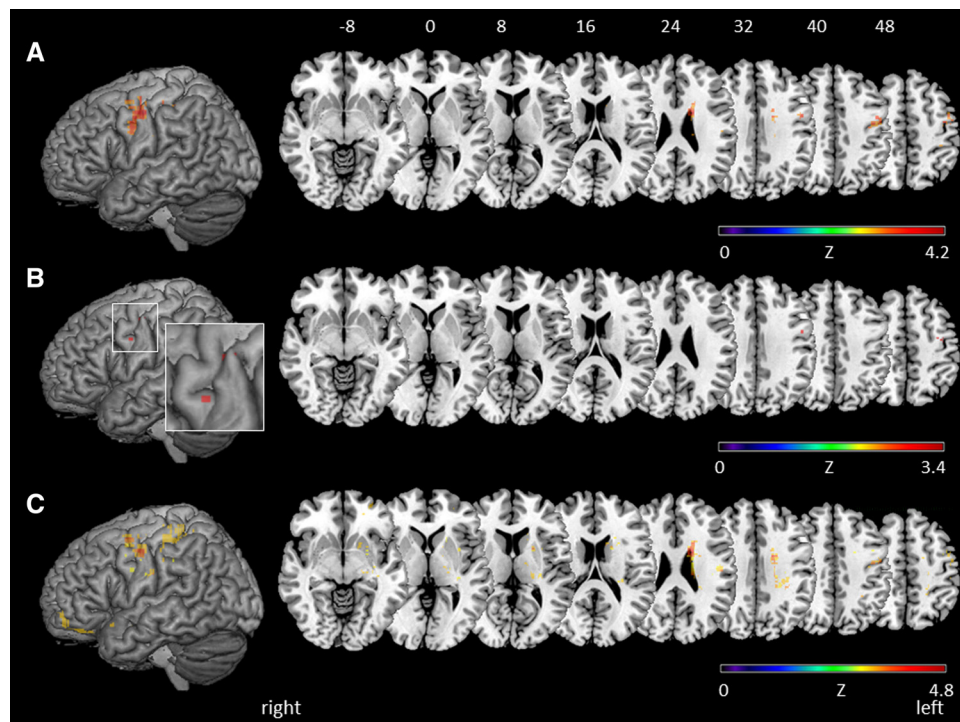


Figure 2. Lesion comparisons between patient groups with apraxia of speech (AOS) and without. The color scale indicates the Z score. **A**, All AOS vs non-AOS group. The threshold for statistical significance was $Z=2.838$ (permutation family-wise error rate [FWE], $P=0.05$), and the maximum voxel value was $Z=4.247$. **B**, Pure AOS vs non-AOS group. The threshold for statistical significance was $Z=2.995$ (permutation FWE, $P=0.05$) and the maximum voxel value was $Z=3.482$. **C**, AOS-aphasia vs non-AOS group. The threshold for statistical significance was $Z=3.005$ (permutation FWE, $P=0.05$), and the maximum voxel value was $Z=4.806$.

Similarly, the posterior wall of the left precentral gyrus predicted the presence of AOS in the comparison between pure AOS and non-AOS ($Z \geq 2.995$, Figure 2B). The comparison between AOS-aphasia and non-AOS indicated scattered lesions, including the basal ganglia, corona radiata, centrum semiovale, and precentral gyrus in the left hemisphere ($Z \geq 3.005$, Figure 2C).

In the comparison between pure AOS and AOS-aphasia, no brain regions associated with pure AOS were detected (Figure 3A). However, scattered subcortical brain regions, including the basal ganglia and corona radiata, were detected in association with AOS-aphasia ($Z \geq 2.108$, Figure 3B).

Discussion

AOS is a motor speech disorder that is clinically characterized by the combination of phonemic segmental changes and articulatory distortions. AOS has been thought to arise from impairment in motor speech planning/programming and differentiated from both aphasia and dysarthria.⁴ However, neuroanatomical foundation of AOS remains controversial. Hillis et al¹⁸ demonstrated the relationship between AOS and damage to the left posterior inferior frontal gyrus, the classic Broca area, in a relatively large number of patients with hyper-acute stroke. In contrast, in his first ever clinicoanatomic investigation of pure AOS (phonetic disintegration syndrome in his terminology), Lecours¹² suggested that damage to the left precentral gyrus caused AOS without a lesion in the inferior frontal gyrus. In agreement with this, a recent lesion-symptom mapping study of patients with chronic stroke by Basilakos

et al²³ demonstrated that lesions in the left precentral gyrus and adjacent somatosensory cortex were predictive of AOS. Moreover, Dronkers²⁰ claimed that AOS was associated with the anterior insula based on their lesion overlapping findings in 25 chronic AOS patients with variable aphasic symptoms. Several factors may be associated with the inconsistent results among the above-mentioned studies on lesional correlates of AOS. First, the time at which symptoms are assessed greatly affects the results. During the chronic stage, the relationship between lesions and symptoms may drastically differ from earlier periods because of spontaneous symptom improvement. Patients who exhibited mild AOS transiently in acute/subacute phase can be classified to non-AOS in chronic stage. Second, the method used for symptom-lesion analysis is critical. When a simple lesion-overlapping method is used for the analysis of patients with stroke, the results may be seriously biased by vascular supply patterns. As previously pointed out, Dronkers seminal study that suggested a relationship between AOS and the anterior insula may have this bias because the anterior insula is one of the most common regions affected by middle cerebral artery territory infarction.^{18,20,32} Third, the precision of lesion localization depends on the modality of brain imaging used. Gyrus identification in adjacent cortical areas is often difficult on perfusion images, including perfusion-weighted images and arterial spin labeling.^{2,18,19}

This study demonstrated that the presence of concomitant aphasia has a substantial impact on the results of lesion analysis for AOS. The regions associated with pure AOS were confined to the left precentral gyrus in the VLSM

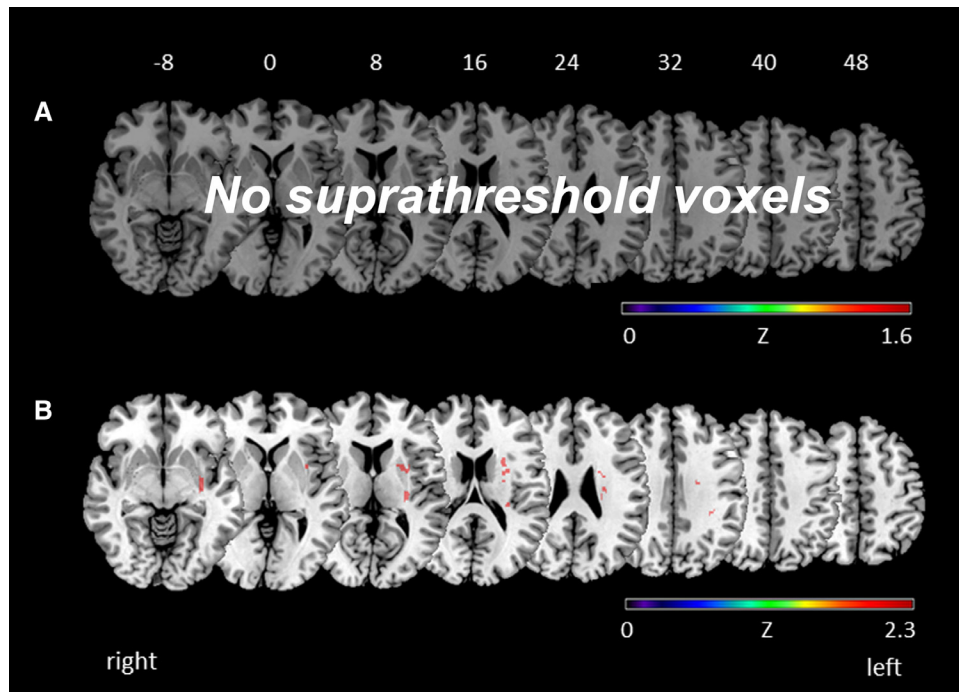


Figure 3. Lesion comparisons between pure apraxia of speech (AOS) and AOS-aphasia groups. **A**, Pure AOS vs AOS-aphasia. The threshold of statistical significance was $Z=2.593$ (permutation family-wise error rate [FWE], $P=0.05$) and the maximum voxel value was $Z=1.671$. **B**, AOS-aphasia vs Pure AOS. The threshold of statistical significance was $Z=2.108$ (permutation FWE, $P=0.05$), and the maximum voxel value was $Z=2.376$.

analysis (Figure 2B; and so also in the lesion overlap map as indicated in Figure 1B), whereas more scattered brain regions were found in the VLSM analysis for AOS-aphasia (Figures 2C and 3B). To our knowledge, most previous case reports of AOS associated with lesions in the left precentral gyrus included patients diagnosed with pure AOS or AOS with mild aphasia.^{12–17} Similarly, a recent lesion-overlapping study by Graff-Radford et al²² demonstrated that the left precentral gyrus and adjacent premotor cortex were the regions of greatest overlap in 7 cases of pure AOS. In contrast, Dronkers²⁰ study, in which patients with both AOS and aphasia accounted for >90% of the total patients with AOS, concluded that this symptom was attributable to lesions in the brain regions outside of the precentral gyrus. A similar discrepancy was found in previous studies on neurodegenerative speech disorders; patients with dominant AOS had atrophy centered in the left precentral gyrus and premotor cortices, whereas those with AOS and aphasia had atrophy extending into Broca area.³³ We suggest that 2 factors may contribute to the variability observed in lesional correlates of AOS with aphasia. First, **patients who have both AOS and aphasia tend to have larger size of lesions and greater involvement of subcortical structures compared with patients with pure AOS** (Figure 1 in this study).^{20,22,23} In such cases, VLSM analysis is prone to detect several or more brain areas in association with a symptom of interest because lesions in different anatomic structures that belongs to the single functional network cause same or similar functional deficits.³⁴ Second, the diagnosis of AOS is more problematic in patients with both AOS and aphasia compared with pure AOS because phonemic paraphasia shares several speech features with AOS.^{7,8}

The diagnosis of AOS is further complicated with additional dysarthria in patients with subcortical lesions. This problem may be improved by using quantitative assessment tools for motor speech abilities.^{7,23}

There are several limitations in this study. First, although data were collected in a standard, preplanned fashion, the speech-language assessments were not performed in a systematic way, particularly for patients without speech deficits or aphasia. Second, the assessment of motor speech abilities in this study was qualitative, and its inter-rater reliability was not examined. Standardized quantitative measures for motor speech abilities should be incorporated in future studies.^{7,23} Third, we may have underestimated the spatial extent of dysfunction because dysfunctional area often extends beyond the regions of infarction observed on T2WI or FLAIR in patients with acute stroke.¹⁸ Diffusion-weighted image would be superior for detecting such areas in acute phase. However, the use of diffusion-weighted image acquired ≈ 9 days after stroke onset can be problematic because of pseudonormalization in apparent diffusion coefficient.

Summary/Conclusions

This VLSM study demonstrated that damage to the left precentral gyrus is critical for the development of AOS in patients with acute stroke. We think that our results are complementary to a recent lesion-symptom mapping study of AOS in patients with chronic stroke.²³

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References

1. Dronkers NF, Plaisant O, Iba-Zizen MT, Cabanis EA. Paul Broca's historic cases: high resolution MR imaging of the brains of Leborgne and Lelong. *Brain*. 2007;130(pt 5):1432–1441. doi: 10.1093/brain/awm042.
2. Richardson JD, Fillmore P, Rorden C, Lapointe LL, Fridriksson J. Re-establishing Broca's initial findings. *Brain Lang*. 2012;123:125–130. doi: 10.1016/j.bandl.2012.08.007.
3. Mohr JP, Pessin MS, Finkelstein S, Funkenstein HH, Duncan GW, Davis KR. Broca aphasia: pathologic and clinical. *Neurology*. 1978;28:311–324.
4. Ogar J, Slama H, Dronkers N, Amici S, Gorno-Tempini ML. Apraxia of speech: an overview. *Neurocase*. 2005;11:427–432. doi: 10.1080/13554790500263529.
5. Josephs KA, Duffy JR, Strand EA, Machulda MM, Senjem ML, Gunter JL, et al. The evolution of primary progressive apraxia of speech. *Brain*. 2014;137(pt 10):2783–2795. doi: 10.1093/brain/awu223.
6. Josephs KA, Duffy JR, Strand EA, Machulda MM, Senjem ML, Master AV, et al. Characterizing a neurodegenerative syndrome: primary progressive apraxia of speech. *Brain*. 2012;135(pt 5):1522–1536. doi: 10.1093/brain/awu032.
7. Strand EA, Duffy JR, Clark HM, Josephs K. The Apraxia of Speech Rating Scale: a tool for diagnosis and description of apraxia of speech. *J Commun Disord*. 2014;51:43–50. doi: 10.1016/j.jcomdis.2014.06.008.
8. Haley KL, Jacks A, de Riesthal M, Abou-Khalil R, Roth HL. Toward a quantitative basis for assessment and diagnosis of apraxia of speech. *J Speech Lang Hear Res*. 2012;55:S1502–S1517. doi: 10.1044/1092-4388(2012/11-0318).
9. Schiff HB, Alexander MP, Naeser MA, Galaburda AM. Aphemia. Clinical-anatomic correlations. *Arch Neurol*. 1983;40:720–727.
10. Shuren J. Insula and aphasia. *J Neurol*. 1993;240:216–218.
11. Nagao M, Takeda K, Komori T, Isozaki E, Hirai S. Apraxia of speech associated with an infarct in the precentral gyrus of the insula. *Neuroradiology*. 1999;41:356–357.
12. Lecours AR. The "Pure Form" of the phonetic disintegration syndrome (pure anarthria); anatomo-clinical report of a historical case. *Brain Lang*. 1976;3:88–113.
13. Tonkonogy J, Goodglass H. Language function, foot of the third frontal gyrus, and rolandic operculum. *Arch Neurol*. 1981;38:486–490.
14. Mori E, Yamadori A, Furumoto M. Left precentral gyrus and Broca's aphasia: a clinicopathologic study. *Neurology*. 1989;39:51–54.
15. Rheims S, Nighoghossian N, Hermier M, Formaglio M, Cakmak S, Derex L, et al. Aphemia related to a premotor cortex infarction. *Eur Neurol*. 2006;55:225–226. doi: 10.1159/000093874.
16. Terao Y, Ugawa Y, Yamamoto T, Sakurai Y, Masumoto T, Abe O, et al. Primary face motor area as the motor representation of articulation. *J Neurol*. 2007;254:442–447. doi: 10.1007/s00415-006-0385-7.
17. Ottomeyer C, Reuter B, Jäger T, Rossmanith C, Hennerici MG, Szabo K. Aphemia: an isolated disorder of speech associated with an ischemic lesion of the left precentral gyrus. *J Neurol*. 2009;256:1166–1168. doi: 10.1007/s00415-009-5055-0.
18. Hillis AE, Work M, Barker PB, Jacobs MA, Breese EL, Maurer K. Re-examining the brain regions crucial for orchestrating speech articulation. *Brain*. 2004;127(pt 7):1479–1487. doi: 10.1093/brain/awh172.
19. Ochfeld E, Newhart M, Molitoris J, Leigh R, Cloutman L, Davis C, et al. Ischemia in broca area is associated with broca aphasia more reliably in acute than in chronic stroke. *Stroke*. 2010;41:325–330. doi: 10.1161/STROKEAHA.109.570374.
20. Dronkers NF. A new brain region for coordinating speech articulation. *Nature*. 1996;384:159–161. doi: 10.1038/384159a0.
21. Ogar J, Willock S, Baldo J, Wilkins D, Ludy C, Dronkers N. Clinical and anatomical correlates of apraxia of speech. *Brain Lang*. 2006;97:343–350. doi: 10.1016/j.bandl.2006.01.008.
22. Graff-Radford J, Jones DT, Strand EA, Rabinstein AA, Duffy JR, Josephs KA. The neuroanatomy of pure apraxia of speech in stroke. *Brain Lang*. 2014;129:43–46. doi: 10.1016/j.bandl.2014.01.004.
23. Basilakos A, Rorden C, Bonilha L, Moser D, Fridriksson J. Patterns of poststroke brain damage that predict speech production errors in apraxia of speech and aphasia dissociate. *Stroke*. 2015;46:1561–1566. doi: 10.1161/STROKEAHA.115.009211.
24. Brott T, Adams HP Jr, Olinger CP, Marler JR, Barsan WG, Biller J, et al. Measurements of acute cerebral infarction: a clinical examination scale. *Stroke*. 1989;20:864–870.
25. SLTA Committee. *Standard Language Test of Aphasia: Manual of Directions*. 2nd ed. Tokyo, Japan: Homeido; 1977.
26. Sasanuma S. Agrammatism in Japanese—A cross-language approach. *J Neurolinguistics*. 1989;4:233–242.
27. Obler LK, Menn L. Agrammatism—The current issues. *J Neurolinguistics*. 1989;3:63–76.
28. Brett M, Leff AP, Rorden C, Ashburner J. Spatial normalization of brain images with focal lesions using cost function masking. *Neuroimage*. 2001;14:486–500. doi: 10.1006/nimg.2001.0845.
29. Rorden C, Karnath HO, Bonilha L. Improving lesion-symptom mapping. *J Cogn Neurosci*. 2007;19:1081–1088. doi: 10.1162/jocn.2007.19.7.1081.
30. Tzourio-Mazoyer N, Landeau B, Papathanassiou D, Crivello F, Etard O, Delcroix N, et al. Automated anatomical labeling of activations in SPM using a macroscopic anatomical parcellation of the MNI MRI single-subject brain. *Neuroimage*. 2002;15:273–289. doi: 10.1006/nimg.2001.0978.
31. Eickhoff SB, Stephan KE, Mohlberg H, Grefkes C, Fink GR, Amunts K, et al. A new SPM toolbox for combining probabilistic cytoarchitectonic maps and functional imaging data. *Neuroimage*. 2005;25:1325–1335. doi: 10.1016/j.neuroimage.2004.12.034.
32. Caviness VS, Makris N, Montinaro E, Sahin NT, Bates JF, Schwamm L, et al. Anatomy of stroke, Part I: an MRI-based topographic and volumetric System of analysis. *Stroke*. 2002;33:2549–2556.
33. Josephs KA, Duffy JR, Strand EA, Whitwell JL, Layton KF, Parisi JE, et al. Clinicopathological and imaging correlates of progressive aphasia and apraxia of speech. *Brain*. 2006;129(pt 6):1385–1398. doi: 10.1093/brain/awl078.
34. Mah YH, Husain M, Rees G, Nachev P. Human brain lesion-deficit inference remapped. *Brain*. 2014;137(pt 9):2522–2531. doi: 10.1093/brain/awu164.

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